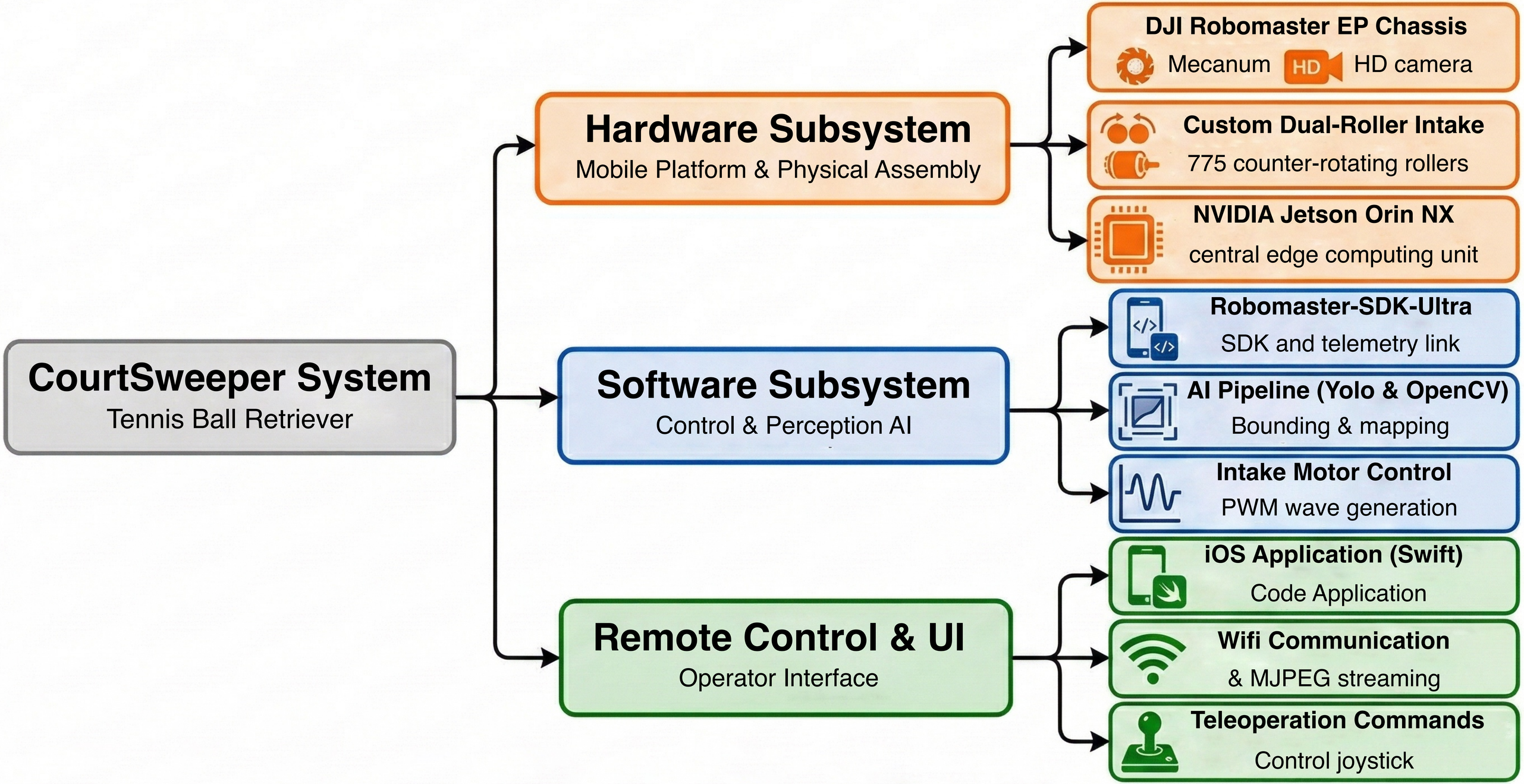
CourtSweeper System
Tennis Ball Retriever
Hardware Subsystem
Mobile Platform & Physical Assembly
DJI Robomaster EP Chassis
Mecanum
HD
HD camera
Custom Dual-Roller Intake
775 counter-rotating rollers
NVIDIA Jetson Orin NX
central edge computing unit
Software Subsystem
Control & Perception AI
Robomaster-SDK-Ultra
SDK and telemetry link
AI Pipeline (Yolo & OpenCV)
Bounding & mapping
Intake Motor Control
PWM wave generation
Remote Control & UI
Operator Interface
iOS Application (Swift)
Code Application
Wifi Communication
& MJPEG streaming
Teleoperation Commands
Control joystick